

DP-700

Microsoft Fabric Data Engineer Associate

Sequential Study Guide • Microsoft Learn Official Materials • Updated April 2026

Pass score: 700/1000 • 3 equal domains (30–35% each) • Intermediate level

Exam Overview

As a Fabric Data Engineer, you should have subject matter expertise with data loading patterns, data architectures, and orchestration processes. This exam tests your ability to:

- Ingest and transform data
- Secure and manage an analytics solution
- Monitor and optimize an analytics solution

You should be skilled at manipulating and transforming data by using SQL, PySpark, and KQL, and will work closely with analytics engineers, architects, analysts, and administrators.

Prerequisites — What you need before starting

- Experience with SQL, PySpark, or Kusto Query Language (KQL)
- Background in data extraction, transformation, and loading (ETL)
- Familiarity with data architectures and orchestration processes
- Experience with enterprise-scale data analytics solutions

Exam Weight Distribution

Technology	Key usage in DP-700
Domain 1	Implement and manage an analytics solution — 30–35%
Domain 2	Ingest and transform data — 30–35%
Domain 3	Monitor and optimize an analytics solution — 30–35%

Step 1 — Foundations: Get started with Microsoft Fabric

Level: Beginner • 10 modules • Start here even if you have some Fabric experience.

URL: learn.microsoft.com/training/paths/get-started-fabric/

Module	What you learn
Intro to end-to-end analytics	Overview of the Fabric platform: what it is, how it works, and its end-to-end analytics capabilities across all workloads.
Get started with lakehouses	Lakehouse architecture: combines data lake storage flexibility with data warehouse analytics on a single SaaS platform.
Use Apache Spark	Spark clusters in Fabric for large-scale data processing and analysis within a lakehouse.
Work with Delta Lake tables	Delta format tables for ACID transactions, time travel, and advanced analytics patterns in the lakehouse.
Orchestrate with pipelines	Data Factory pipelines for orchestrating data ingestion and transformation tasks.
Ingest with Dataflows Gen2	Power Query Online for visual, no-code multi-step data ingestion and transformation.
Get started with data warehouses	Relational data warehouses in Fabric using T-SQL for structured analytics workloads.
Real-Time Intelligence intro	Ingesting, querying, and processing real-time data streams using KQL and Eventhouse.
Data Science in Fabric	Managing notebooks, experiments, and ML models within the Fabric ecosystem.
Administer Fabric environment	Configure features and manage access as an administrator on the SaaS platform.

Step 2 — Implement a Lakehouse with Microsoft Fabric

Level: Intermediate • 7 modules • Prerequisite: Step 1 complete.

URL: learn.microsoft.com/training/paths/implement-lakehouse-microsoft-fabric/

Module	What you learn
Intro to end-to-end analytics	Revisit Fabric architecture from an implementation lens; understand workload integration.
Get started with lakehouses	Create and manage lakehouses; understand storage layers and SQL analytics endpoints.
Use Apache Spark	Process and analyze data at scale using Spark notebooks with PySpark.
Work with Delta Lake tables	Build reliable, queryable Delta tables; use V-Order compaction and time travel.
Ingest with Dataflows Gen2	Visually ingest and transform multi-source data into the lakehouse using Power Query.
Orchestrate processes & pipelines	Build pipelines with triggers, parameters, and dynamic expressions for orchestration.
Medallion architecture design	Organize data across Bronze → Silver → Gold layers for optimized, incremental analytics.

Step 3 — Ingest Data with Microsoft Fabric

Level: Intermediate • 6 modules • Prerequisite: Steps 1–2 complete.

URL: learn.microsoft.com/training/paths/ingest-data-with-microsoft-fabric/

Module	What you learn
Dataflows Gen2	Multi-step visual ingestion and transformation using Power Query Online; schedule and publish.
Pipelines & orchestration	Design full and incremental load pipelines; implement event-based triggers and scheduling.
Apache Spark in Fabric	PySpark for large-scale batch transformation, aggregation, denormalization, and deduplication.
Real-Time Intelligence intro	Ingest and query streaming data using KQL, Eventhouse, and Real-Time hub.
Real-time Eventstreams	No-code ingestion and transformation of real-time events from multiple sources with routing.
Work with Eventhouse	Scalable store for real-time data; KQL queries, windowing functions, and OneLake shortcuts.

Step 4 — Manage a Microsoft Fabric Environment

Level: Intermediate • 4 modules • Prerequisite: Steps 1–3 complete.

URL: learn.microsoft.com/training/paths/manage-microsoft-fabric-environment/

Module	What you learn
CI/CD in Microsoft Fabric	Git integration and deployment pipelines for version control, database projects, and lifecycle management.
Monitor Fabric activities	Monitoring Hub, configure alerts, and Activator for detecting patterns in streaming data.
Secure data access	Multi-layer security: workspace, item, row, column, object, folder/file, and OneLake security.
Administer Fabric environment	Configure tenant settings, manage capacity, domains, workspace settings, and audit logs.

Step 5 — Complete Exam Skill Checklist (All 3 Domains)

Each domain carries equal weight. Use this table as a revision checklist before your exam.

Domain 1: Implement & Manage (30-35%)	Domain 2: Ingest & Transform (30-35%)	Domain 3: Monitor & Optimize (30-35%)
- Configure Spark workspace settings - Configure domain & OneLake workspace settings - Configure Dataflows Gen2 workspace settings - Configure version control (Git integration) - Implement database projects - Create & configure deployment pipelines - Workspace-level access controls - Item-level access controls - Row, column, object & folder/file-level security - Dynamic data masking - Apply sensitivity labels to items - Endorse items - Implement & use Fabric audit logs - Configure & implement OneLake security - Choose between Dataflow Gen2, pipeline & notebook - Design schedules & event-based triggers - Orchestration with parameters & dynamic expressions	- Design full & incremental data loads - Prepare data for dimensional model loading - Design streaming data load patterns - Choose appropriate data store - Choose Dataflows Gen2, notebooks, KQL or T-SQL - Create & manage OneLake shortcuts - Implement mirroring - Ingest data by using pipelines - Transform data with PySpark, SQL & KQL - Denormalize data - Group & aggregate data - Handle duplicates, missing & late-arriving data - Choose streaming engine - Native tables vs OneLake shortcuts in RTI - Query acceleration vs standard OneLake shortcuts - Process data using Eventstreams - Process data using Spark structured streaming - Process data using KQL - Create windowing functions	- Monitor data ingestion - Monitor data transformation - Monitor semantic model refresh - Configure alerts - Resolve pipeline errors - Resolve Dataflow Gen2 errors - Resolve notebook errors - Resolve Eventhouse errors - Resolve Eventstream errors - Resolve T-SQL errors - Resolve OneLake shortcut errors - Optimize Lakehouse tables - Optimize pipelines - Optimize data warehouses - Optimize Eventstreams & Eventhouses - Optimize Spark performance - Optimize query performance

Step 6 — Key Technologies to Master

Technology	Key usage in DP-700
PySpark	Batch transforms, large-scale data processing, aggregation, Spark structured streaming for real-time
T-SQL	Warehouse queries, transformations, row-level & column-level security, performance optimization
KQL	Real-time queries, Eventhouse analytics, windowing functions, data exploration
Delta Lake	ACID transactions, time travel, V-Order compaction, medallion architecture (Bronze/Silver/Gold)

Technology	Key usage in DP-700
Dataflows Gen2	No-code/low-code visual data ingestion and transformation using Power Query Online
Eventstreams	Real-time event ingestion, routing, filtering, and transformation from multiple sources
Pipelines	Orchestration of data movement, scheduling, event triggers, parameters, and dynamic expressions
OneLake	Unified storage layer; shortcuts, mirroring, folder/file-level security, cross-workspace access

Exam Preparation Tips

Practical tips for passing DP-700
• Free practice assessment available at learn.microsoft.com — take it before and after studying to measure gaps.
• Each module includes hands-on exercises. Use a free Microsoft Fabric trial to complete labs.
• Score of 700 out of 1000 required to pass. Most questions are scenario-based, not purely factual.
• Exam is typically 45–65 questions; scenario questions test judgment (e.g., when to use Dataflow Gen2 vs pipeline vs notebook).
• Certification expires after 1 year — renew free via an online assessment on Microsoft Learn.
• All three domains carry equal weight — do not skip monitoring & optimization.
• Focus on real-world decision-making: choosing tools, troubleshooting errors, and optimizing performance.

Official Resources & Links

- Exam page: learn.microsoft.com/credentials/certifications/fabric-data-engineer-associate/
- Study guide: learn.microsoft.com/credentials/certifications/resources/study-guides/dp-700
- Free practice assessment: search 'DP-700 practice assessment' on learn.microsoft.com
- Fabric documentation: learn.microsoft.com/fabric/
- Data Engineering overview: learn.microsoft.com/fabric/data-engineering/data-engineering-overview
- Community: techcommunity.microsoft.com > Analytics on Azure
- Exam sandbox (preview the exam UI): aka.ms/examdemo

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